# Solar Energy Management

User Manual

Intelligent solar optimization, EV charging, battery management,
and peak load control — all in one integration.

Version 2.7 · Home Assistant 2025.3+

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# 1. Installation

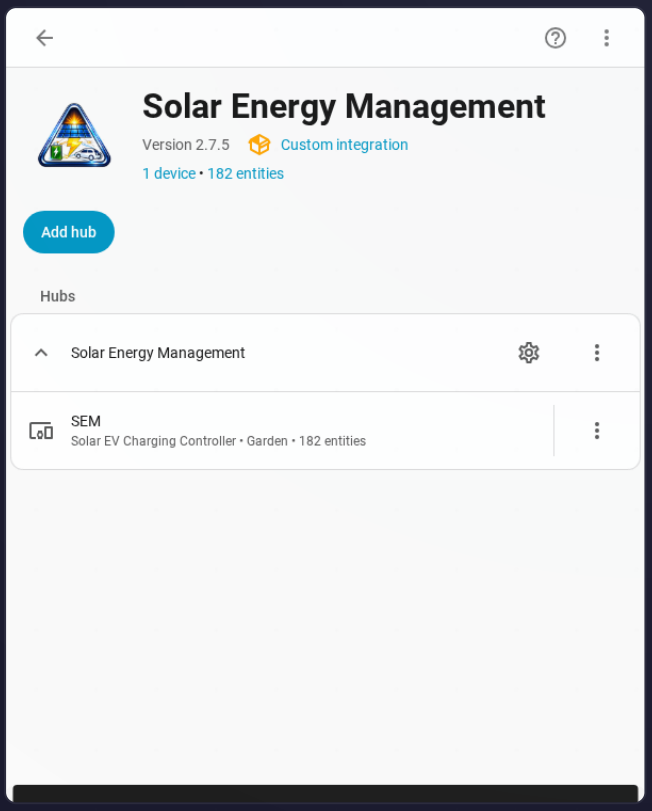
SEM is installed via HACS or manually. Setup takes about 5 minutes.

## Prerequisites

- **Home Assistant 2025.3+**
- **HA Energy Dashboard** configured (solar + grid sensors)
- **EV charger** with HA integration (KEBA, Wallbox, go-eCharger, Easee)
- **5 HACS frontend cards:** mushroom, card-mod, apexcharts-card, sankey-chart, fold-entity-row

## Install via HACS

1. Open **HACS > Integrations**
2. Click 3-dot menu > **Custom repositories**
3. URL: `https://github.com/belinea4071/solar_energy_management`
4. Search "Solar Energy Management" > **Download**
5. **Restart Home Assistant**



# 2. Setup Wizard

Go to **Settings > Devices & Services > Add Integration** and search "Solar Energy Management".

## Step 1: Energy Dashboard Detection

SEM auto-reads your solar, grid, and battery sensors from the HA Energy Dashboard. A summary is shown. Optionally enable Observer Mode for read-only monitoring.

## Step 2: EV Charger

| Field | Description |
|---|---|
| Connected sensor | Binary sensor: is the car plugged in? |
| Charging sensor | Binary sensor: is charging active? |
| Power sensor | Current charging power (W) |

## Step 3: Hardware

| Field | Default | Description |
|---|---|---|
| Battery capacity | 10 kWh | Home battery size |
| Target peak limit | 5 kW | 15-min rolling peak target |
| Generate dashboard | ON | Auto-create 7-tab dashboard |

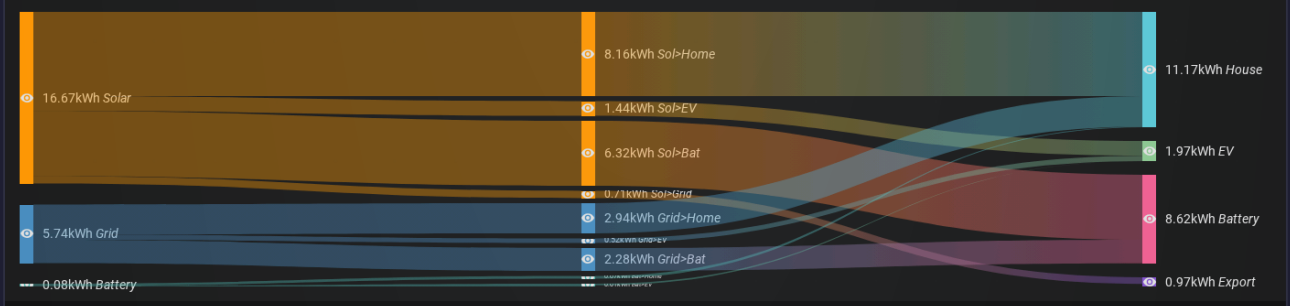
Click **Submit** — done! After a restart the dashboard appears in the sidebar.

# 3. Dashboard

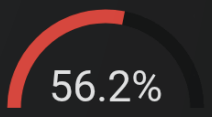
SEM generates a 7-tab dashboard with a unified glassmorphism dark theme.

| Tab | Content |
|---|---|
| **Home** | System diagram, solar summary, 7-day chart, weather, recommendations |
| **Energy** | Sankey, self-consumption/autarky gauges, donut, 24h power, $CO_2$, trees |
| **Battery** | SOC gauge, charge/discharge chart, zone configuration |
| **EV** | Charging status, session stats, solar share, lifetime statistics |
| **Control** | Device priority, EV/battery/tariff settings |
| **Costs** | Today/month/year KPIs, charts, demand charge, tariff rates |
| **System** | Diagnostics, sensor status, configuration |

## Energy Tab

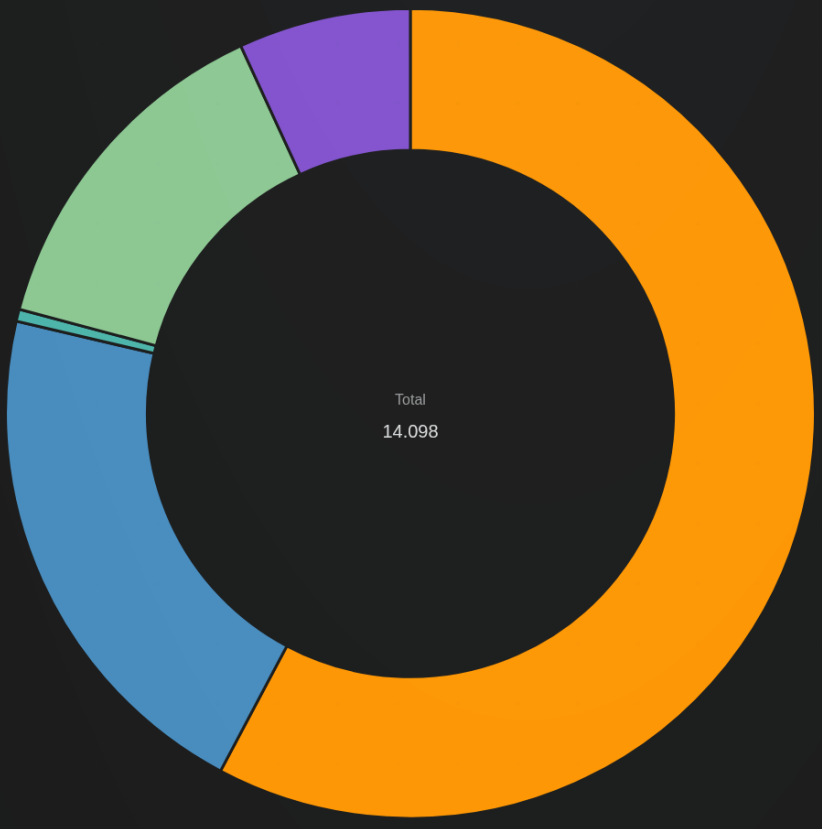


Self-consumption



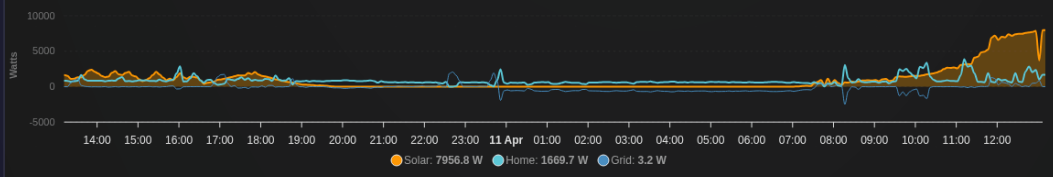

Autarky

Today's Energy Distribution



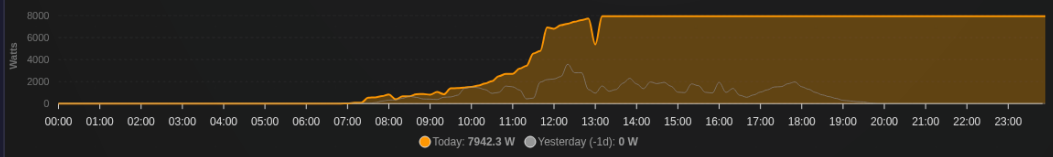

Solar > Home: **8.1 kWh** Grid > Home: **2.9 kWh** Battery > Home: **0.1 kWh** EV Charging: **2 kWh** Export: **1 kWh**

24h Power




Solar: **7956.8 W** Home: **1669.7 W** Grid: **3.2 W**

Solar — Today vs Yesterday

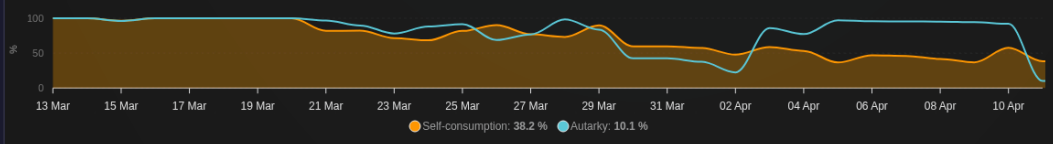



Today: **7942.3 W** Yesterday (-1d): **0 W**

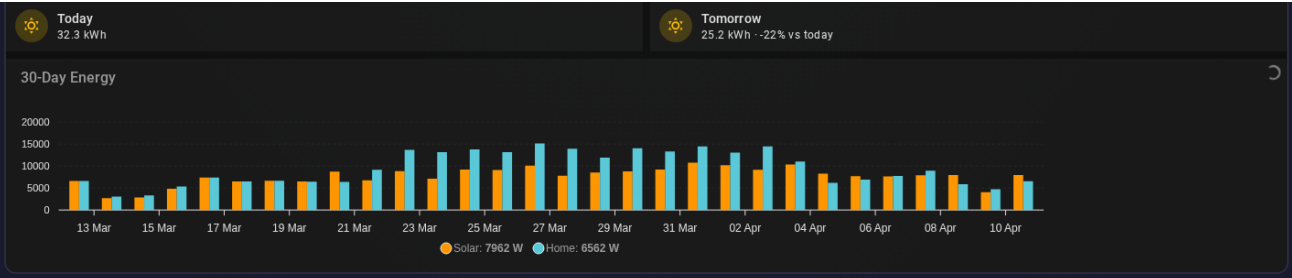
**Carbon Avoided Today**
1.2 kg CO2 (9.6 kWh self-consumed × 128 g/kWh)

**0.1 Trees Equivalent**
2.0 kg $CO_2$ avoided this year · Lifetime: 2 kg

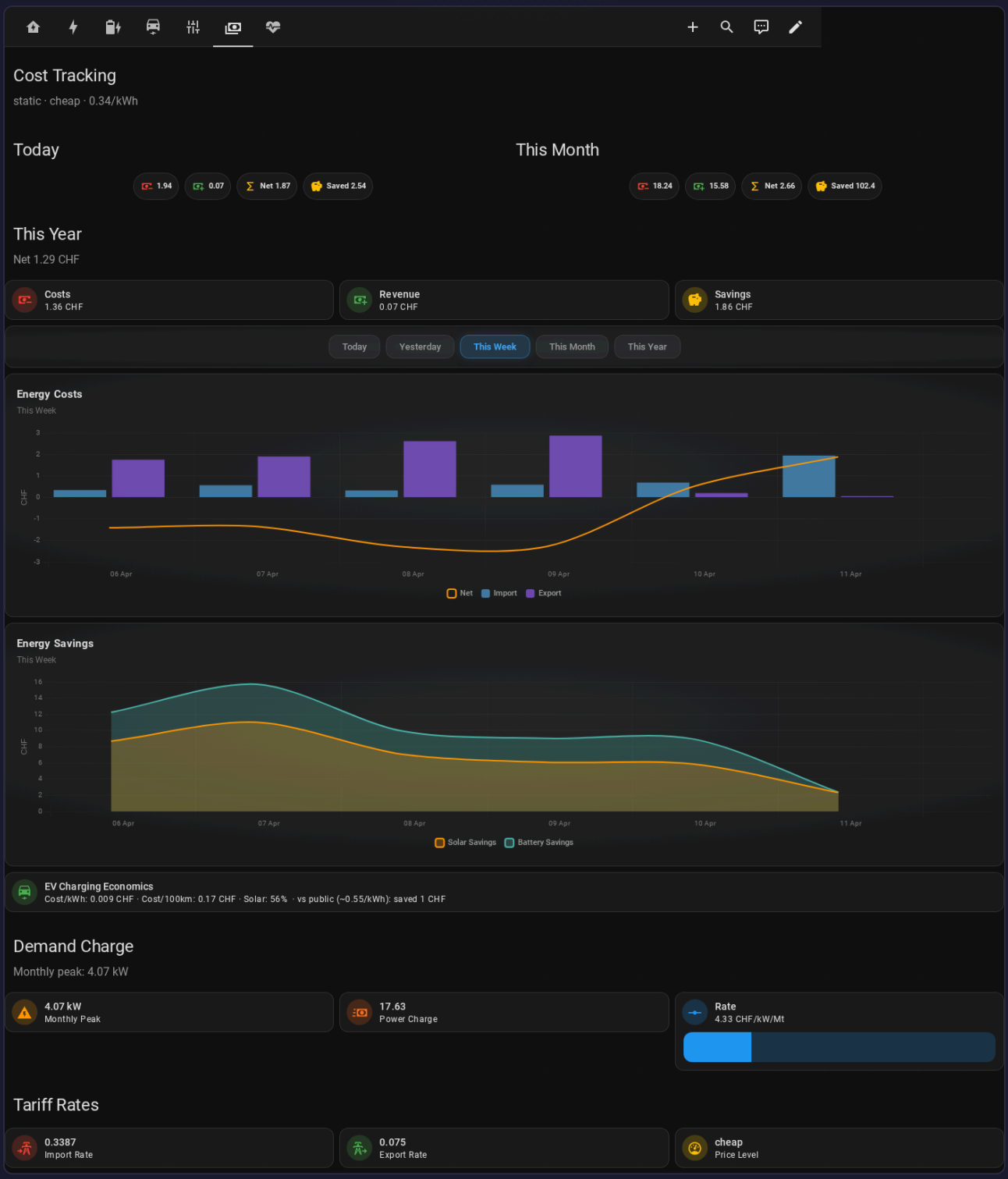
Self-Consumption — 30 Day Trend




Self-consumption: **38.2 %** Autarky: **10.1 %**


Today
32.3 kWh
Tomorrow
25.2 kWh · -22% vs today
30-Day Energy
20000
15000
10000
5000
0
13 Mar
15 Mar
17 Mar
19 Mar
21 Mar
23 Mar
25 Mar
27 Mar
29 Mar
31 Mar
02 Apr
04 Apr
06 Apr
08 Apr
10 Apr
Solar: 7962 W
Home: 6562 W

## Costs Tab


Cost Tracking
static · cheap · 0.34/kWh
Today
1.94
0.07
Net 1.87
Saved 2.54
This Month
18.24
15.58
Net 2.66
Saved 102.4
This Year
Net 1.29 CHF
Costs
1.36 CHF
Revenue
0.07 CHF
Savings
1.86 CHF
Today
Yesterday
This Week
This Month
This Year
Energy Costs
This Week
CHF
06 Apr
07 Apr
08 Apr
09 Apr
10 Apr
11 Apr
Net
Import
Export
Energy Savings
This Week
CHF
Solar Savings
Battery Savings
EV Charging Economics
Cost/kWh: 0.009 CHF · Cost/100km: 0.17 CHF · Solar: 56% · vs public (~0.55/kWh): saved 1 CHF
Demand Charge
Monthly peak: 4.07 kW
4.07 kW
Monthly Peak
17.63
Power Charge
Rate
4.33 CHF/kW/Mt
Tariff Rates
0.3387
Import Rate
0.075
Export Rate
cheap
Price Level

# 4. Charging Modes

### ☀️ Solar Charging

Dynamically adjusts EV current (6-32A) to match solar surplus. Pauses for clouds, resumes automatically.

### 🌙 Night Charging

Grid charging during off-peak hours. Configurable window. Smart forecast reduction for sunny mornings.

### 🔋 Battery Assist

When battery SOC is high, it supplements solar to maintain EV charging through cloudy periods.

### 👁 Observer Mode

Read-only monitoring — SEM tracks all data but sends no commands to hardware.

## The 3 Switches

| Switch | Default | Purpose |
|---|---|---|
| `switch.sem_night_charging` | ON | Enable overnight grid charging |
| `switch.sem_observer_mode` | OFF | Read-only, no hardware control |
| `switch.sem_forecast_night_reduction` | OFF | Reduce night target when tomorrow is sunny |

Everything else — solar charging, battery protection, peak management — is fully automatic.

## 4b. Battery Strategy

SEM uses a 4-zone SOC model to balance battery and EV:

| Zone | SOC Range | Behavior |
|---|---|---|
| Priority | 0–80% | Battery charges first, EV waits |
| Buffer | 80–70% | EV gets solar, battery-assist enabled |
| Auto-start | 70–30% | Normal solar operation |
| Protection | <20% | EV pauses, battery protected |

## 5. Peak Load Management

SEM tracks the 15-minute rolling peak and sheds loads when the limit is exceeded.

- **Drag-and-drop** device priority on the Control tab
- Highest priority devices are shed first
- **Critical** devices are never shed
- Monthly peak stored for demand charge calculation

## 6. Costs & Environment

SEM tracks costs and savings on a daily, monthly, and yearly basis.

| Sensor | Description |
|---|---|
| Daily costs | Grid import costs, export revenue, net cost, savings |
| Monthly costs | Monthly aggregated financials |
| Yearly costs | Year-to-date totals (reset on Jan 1) |
| $CO_2$ avoided | Daily/yearly/lifetime based on self-consumed solar × 128g/kWh |
| Trees equivalent | $CO_2$ / 22 kg per tree/year (EPA estimate) |

The trees icon grows with impact: 🌱 sprout → 🌳 tree → 🌲 pine → 🌲🌲 forest

## Cost Tracking

static · cheap · 0.34/kWh

### Today

1.94 | 0.07 | Net 1.87 | Saved 2.54

### This Month

18.24 | 15.58 | Net 2.66 | Saved 102.4

### This Year

Net 1.29 CHF

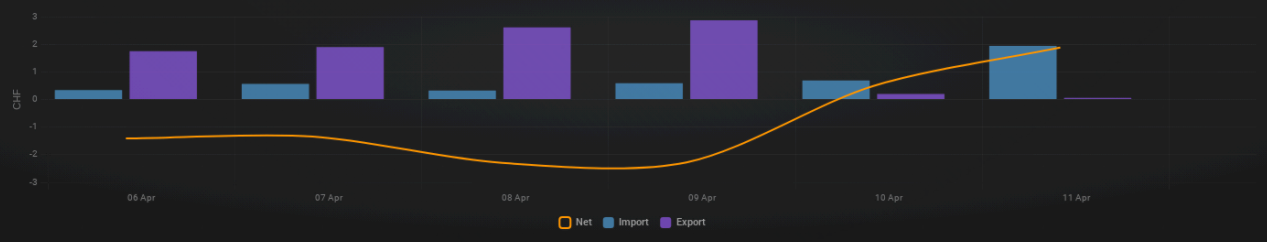
**Costs**
1.36 CHF

**Revenue**
0.07 CHF

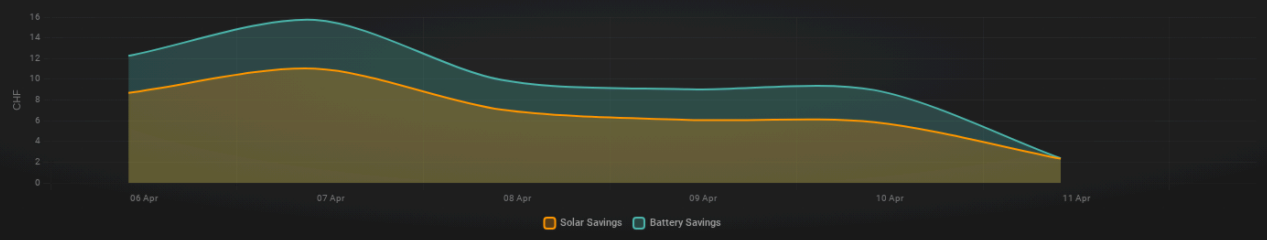
**Savings**
1.86 CHF

Today | Yesterday | This Week | This Month | This Year

**Energy Costs**
This Week




**Energy Savings**
This Week




**EV Charging Economics**
Cost/kWh: 0.009 CHF · Cost/100km: 0.17 CHF · Solar: 56% · vs public (~0.55/kWh): saved 1 CHF

## Demand Charge

Monthly peak: 4.07 kW

**4.07 kW**
Monthly Peak

**17.63**
Power Charge

**Rate**
4.33 CHF/kW/Mt

## Tariff Rates

**0.3387**
Import Rate

**0.075**
Export Rate

**cheap**
Price Level

# 7. Configuration

All settings under **Settings > Devices & Services > Solar Energy Management > Configure**:

| Step | Settings |
|---|---|
| EV Charger | Sensor mappings, charger service |
| SOC Zones | Priority, buffer, auto-start, protection SOC levels |
| EV Charging | Daily target, minimum/night current, phases |
| Tariff & Advanced | Import/export rates, update interval |
| Load Management | Peak limit, shedding margin |
| Notifications | KEBA display, mobile push |

Or adjust directly on the **Control tab** of the dashboard.

## EV Charging

Solar mode - Battery assist · 2.0/10 kWh

**Night Charging**
On

**Forecast Reduction**
On

**Night Charge Target (kWh)**
9.5 kWh

**Night Start (A)**
10 A

**Stall Cooldown (sec)**
120 s

## Surplus Control

9/10 devices · 1184W allocated

**Surplus Available**
3128W total · 3126W distributable

**Allocation**
1184W used · 1943W free

**Regulation Offset (W)**
50 W

## Battery Management

SOC 71% · Charging

**Priority SOC**
30%

**Minimum SOC**
30%

**Resume SOC**
50%

**Capacity (kWh)**
NaN kWh

## Heat Pump & Hot Water

normal · SG-Ready 2

**SG-Ready State**
2

**Mode**
normal

**Boost Offset**
2 °C

**Hot Water Max**
60 °C

## Solar & Power

**Min Solar (W)**
1000 W

**Max Grid Import (W)**
0 W

## Tariff & Pricing

static · cheap

**Current Electricity Price**
0.3387

**Cheap Threshold**
0.35 CHF/kWh

**Expensive Threshold**
0.35 CHF/kWh

## Load Management

Device priority · 11 devices

NORMAL

Peak 0.02 kW / 6.0

| | |
|---|---|
| Current Peak | 0.02 kW |
| Target Limit | 6.00 kW |
| Margin | 5.98 kW |

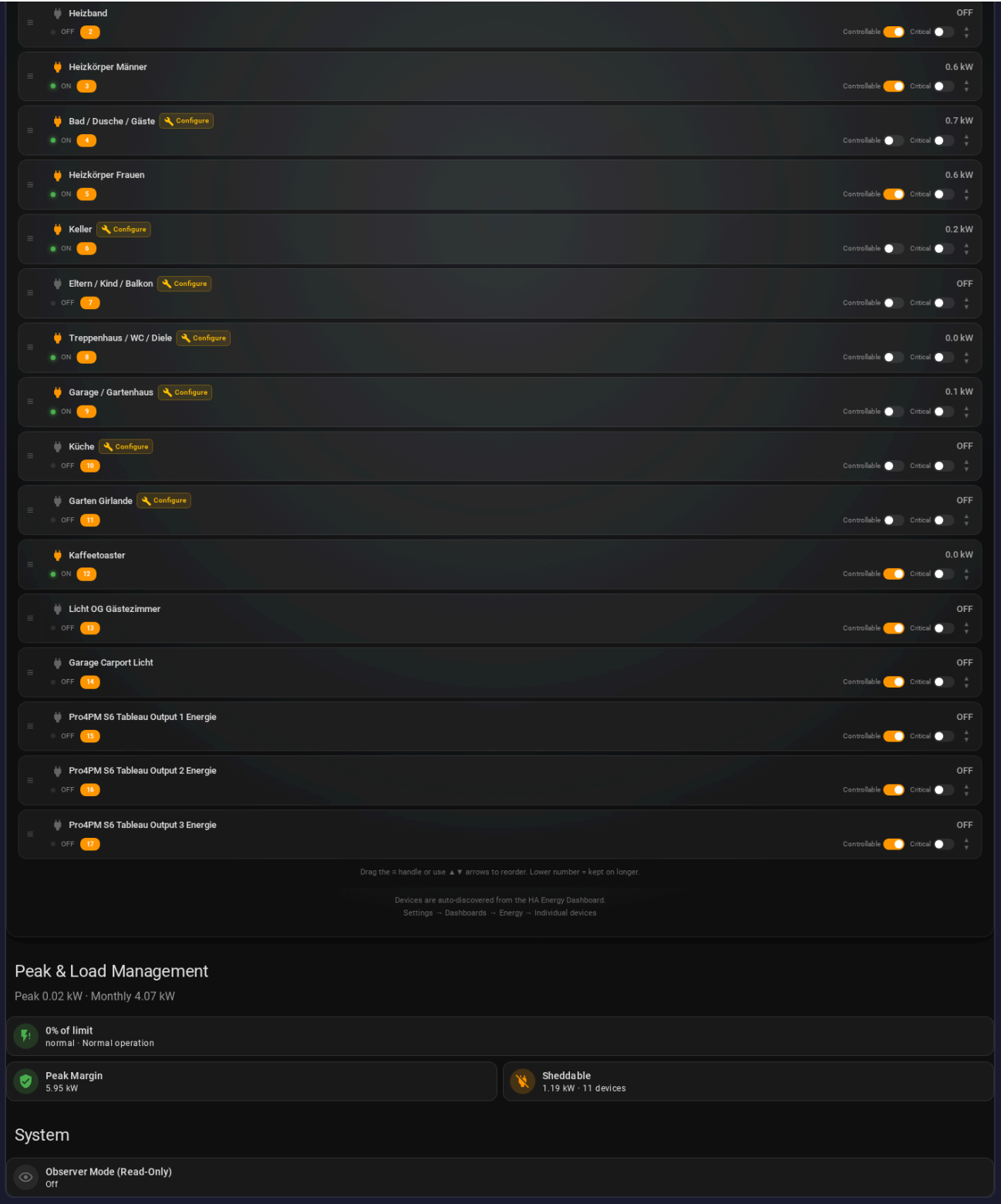
Adjust Target Peak Limit

6 kW Set

Drag to reorder · 1 = highest priority

KEBA P30 OFF

OFF 1 Controllable Critical




# 8. Troubleshooting

| Problem | Solution |
|---|---|
| Sensors "unavailable" | Check Energy Dashboard config, hardware integration online? |
| Dashboard blank | Install `card-mod` HACS card, hard-refresh browser |
| Grid import/export swapped | SEM auto-detects direction via Energy Dashboard counters |
| EV not charging | Check connected sensor, solar surplus > minimum power? |